

ARGUS: Auditing the Identification Credibility of Climate-Policy Causal Evaluations

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Abstract

Difference-in-differences (DID) studies are widely used to evaluate climate policy, yet their identification assumptions are more often *asserted* than *audited*. Because the counterfactual is unobserved, we ask instead whether each assumption is adequately supported by the evidence a paper reports. We introduce ARGUS, an end-to-end, *bounded-agentic* auditing system in which an LLM runs behind a fixed, deterministic pipeline and scores DID studies on an 11-dimension assumption–implication–evidence rubric, retrieving evidence, judging its adequacy, and producing a risk report for expert review rather than adjudicating causal truth. Under realistic flaw injection ARGUS catches all 11 injected flaws versus 5 for a same-model single-pass baseline, with no false alarms; across 27 real papers it exposes evidence grounding as the bottleneck, abstaining when support is absent; and a pilot expert study reveals a systematic over-severity that a lightweight calibration layer partly corrects. ARGUS is thus a high-sensitivity screening tool that surfaces weakly evidenced identification assumptions for expert review.

1 Introduction

Difference-in-differences (DID) is one of the most widely used research designs in empirical economics, and a central tool for evaluating climate and environmental policy. Studies of carbon pricing, emissions-trading pilots, and industrial-emission standards routinely compare treated and untreated units before and after a policy to recover its causal effect. The credibility of any such estimate rests entirely on a chain of *identification assumptions*: that treated and control units would have moved in parallel absent the policy, that units did not anticipate it, that staggered adoption is handled with an appropriate estimator, that there is no interference between

treated and control units, and so on. A decade of methodological work has shown how fragile each of these assumptions can be, and how easily a standard two-way fixed-effects specification can mislead under heterogeneous and staggered treatment timing (Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfœuille, 2020; Roth et al., 2023).

Yet in applied work these assumptions are frequently *asserted* rather than *audited*. A paper states that parallel trends “hold”, shows an event-study plot, and moves on; a reader who wants to know whether the identification argument is actually supported by the reported evidence must reconstruct it by hand, dimension by dimension. This is slow, expert-intensive, and inconsistent across reviewers—a reliability gap in a body of evidence that increasingly informs climate and environmental policy.

Our stance. We argue that identification *credibility* is auditable even though the causal *effect* is not. In a DID design the counterfactual is never observed, so asking an automated system whether the estimated effect is true is both hopeless and the wrong target. We ask instead a question that can be grounded in the paper itself: *is each identification assumption supported by the evidence the study reports, and where is that support weakest?* This reframing turns an unanswerable causal question into a tractable auditing problem over assumptions, their testable implications, and the in-paper evidence that bears on them—localizing where support is fragile rather than issuing a verdict.

Why human calibration. Identification credibility is not a binary truth label. A dimension is rarely simply “satisfied” or “violated”; reviewers grade it on a severity scale, and where the low/medium/high boundaries fall is a matter of ex-

pert judgement. Some disagreement is therefore intrinsic, especially when a DID-specific check is applied to a non-standard design where the dimension may not apply as conventionally defined. We treat this ambiguity as a feature of the task, not noise: ARGUS pairs automated, evidence-grounded auditing with a *human-calibrated severity* scale, using a pilot expert gold both to *evaluate* the auditor and to *calibrate* how severely it flags.

ARGUS. We introduce ARGUS, a responsible-AI framework that audits the identification credibility of DID environmental-policy studies. ARGUS decomposes identification into a fixed set of auditable dimensions, gathers the supporting evidence each dimension requires, assesses every assumption–implication–evidence chain, localizes risk across dimensions, and produces a transparent, evidence-anchored report for a human expert (Figure 1). It is a *bounded-agentic* system by design: the pipeline between modules is fixed and deterministic, and model-driven agency is confined to two scoped sub-tasks. Because the true effect is unavailable as ground truth, we validate the auditor with *synthetic flaw injection*: we perturb a credible paper with one known identification flaw and check whether ARGUS detects it, stays quiet on the clean version, and localizes the right dimension. The injected flaw is local ground truth that requires no access to the true effect. Deploying ARGUS on real papers then exposes *evidence grounding* (retrieving the right evidence) as the binding constraint, and a pilot human gold reveals a systematic over-severity bias that a lightweight, anchor-based calibration layer reduces while preserving high-risk recall. ARGUS is thus best read as a high-sensitivity screening tool that flags dimensions for expert review rather than an autonomous arbiter of causal truth.

Contributions. This paper makes four contributions. (i) a structured **11-dimension audit framework** for DID identification credibility whose unit of audit is an explicit *assumption–implication–evidence* chain, targeting reported-evidence adequacy rather than causal truth (§3); (ii) a **bounded-agentic** evidence-retrieval and assessment pipeline, with model agency confined to two scoped, traceable sub-tasks behind a fixed deterministic control flow; (iii) a **multi-stage evaluation** built on *flaw injection as local ground truth* (it needs no access to the true effect), showing that adequacy reasoning catches flawed-but-

present evidence that keyword presence cannot; a real-paper deployment on 27 DID papers relocates the bottleneck to *evidence grounding* and separates retrieval failure from substantive high risk, and an *architecture ablation* against a same-model single-pass baseline shows the bounded design raises synthetic flaw detection from 0.455 to 1.000 and, on real papers, preserves an explicit unknown state (38.9% vs 0.4%) that keeps evidence-grounding uncertainty visible rather than committing a guess (§5); and (iv) a **pilot human-gold calibration loop** (5 papers, 55 dimension-level cells) in which disagreement is largely design-applicability-driven: comparing ARGUS to an adjudicated gold reveals a one-directional over-severity bias (more severe than gold in 28/33 answered cells, less severe in none), and an applicability-aware, anchor-based calibration layer reduces it while preserving high-risk recall. Expert annotation can therefore both evaluate and calibrate an identification audit; the human-gold analysis is pilot-scale, and larger expert-labelled corpora are future work.

2 Related Work

DID identification and its threats. A large econometrics literature formalizes when DID estimators fail under staggered timing and heterogeneous effects—the “forbidden” comparisons of two-way fixed effects (Goodman-Bacon, 2021), negative weights (de Chaisemartin and D’Haultfœuille, 2020), robust estimators (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021), assumption and pre-trend surveys (Roth et al., 2023), and fragile inference under serial correlation (Bertrand et al., 2004). ARGUS adds no estimator: it *operationalizes* this literature as an auditable rubric, turning each known threat into a dimension whose supporting evidence can be checked in a given paper (§3).

LLM causal reasoning and benchmarks. A parallel line asks whether LLMs reason about causality. Benchmarks (CLadder (Jin et al., 2023), Corr2Cause (Jin et al., 2024), CausalBench (Zhou et al., 2024)) and critiques (Kiciman et al., 2024; Yang et al., 2024; Zečević et al., 2023) find models often reproduce causal *language* without reliable causal *inference*; a few move toward applied settings and toward separating *identification* from *estimation* (Lee et al., 2025; Sawarni et al., 2026). ARGUS inverts the task: rather than producing

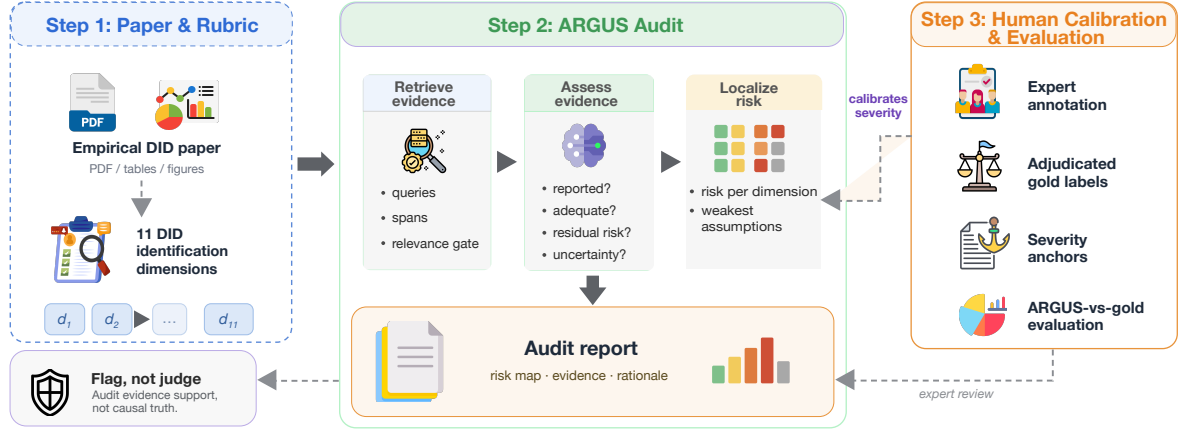


Figure 1: Overview of ARGUS. Step 1 converts an empirical DID paper into an auditable structure using an 11-dimension identification rubric. Step 2 retrieves dimension-specific evidence, assesses reported-evidence adequacy with a bounded LLM audit module, and localizes risk in a human-reviewable report. Step 3 uses expert annotation and adjudicated gold labels to evaluate and calibrate audit severity. ARGUS flags dimensions for expert review; it does not adjudicate causal truth.

a causal answer, it audits whether a *human* paper’s identification assumptions are adequately evidenced, and abstains from a causal verdict by design.

Responsible AI for empirical research. AI is increasingly applied to research reliability—reproducibility checking, statistical-reporting validation, and AI-assisted review—motivated by wide analyst-driven result spreads (Silberzahn et al., 2018; Botvinik-Nezer et al., 2020) and specification searching in causal economics (Brodeur et al., 2020). ARGUS differs in target and posture: it audits *identification credibility* specifically—the part of an argument that licenses a causal claim—and issues no verdicts, producing evidence-anchored, human-reviewable reports that localize risk and leave adjudication to an expert. A complementary line asks whether a model can *carry out* inference—running generated estimation code against synthetic data; ARGUS instead audits the prior layer, whether identification *assumptions* are supported by the evidence a paper reports.

Evaluation methodology and bounded agents. Our evaluation follows benchmarking lessons: holistic evaluation (Liang et al., 2023) over a single number (we report a layered retrieval/relevance/adequacy/risk profile); dynamic, adversarial benchmarking (Kiela et al., 2021) (our model-in-the-loop flaw injection); and the caution that operationalizing a capability is non-neutral (Raji et al., 2021; Gebru et al.,

2021) (§6). Where ReAct-style agents (Yao et al., 2023) act over an open horizon, ARGUS *bounds* the paradigm—agency confined to two scoped sub-tasks behind deterministic control flow—trading autonomy for reproducibility and evidence-traceability.

3 The ARGUS Framework

3.1 Identification rubric

ARGUS decomposes a DID study into eleven auditable dimensions: parallel trends, no anticipation, staggered-timing handling, SUTVA/spillovers, control-group construction, specification, inference, sample-period selection, concurrent policies, placebo/robustness, and data measurement. Each dimension is expressed as an *assumption* (what must hold for identification), a testable *implication* (what should therefore be observable), and the *evidence* a credible paper would report to support it. For a carbon-pricing or emissions-trading evaluation, for instance, the parallel-trends dimension expects an event-study or pre-trend test with null pre-period coefficients, while no-anticipation turns on the gap between a scheme’s *announcement* and *implementation* dates. The rubric is declarative, stored in a configuration file (`identification_dimensions.yaml`), so the audited dimensions are explicit and inspectable rather than buried in code. A companion flaw taxonomy (§4) maps each known identification threat one-to-one onto a dimension, which

both documents the rubric’s coverage and supplies the perturbations used for evaluation.

3.2 Bounded-agentic pipeline

The audit runs a fixed, deterministic sequence—*decomposition* → *extraction* → *assessment* → *localization* → *report*—shown in the top band of Figure 2. The stage order is written in code and never chosen by a model. *Decomposition* instantiates the rubric for the paper at hand; *extraction* gathers, per dimension, the in-paper evidence relevant to that assumption; *assessment* evaluates each assumption–implication–evidence chain and emits a per-dimension risk judgement with a rationale and cited evidence; *localization* aggregates these judgements into a risk map ranking where support is weakest; and *report* renders a transparent, evidence-anchored document for a human reviewer.

Agency is confined to the two stages where open-ended judgement actually helps—extraction and assessment. Each invokes a bounded ReAct-style loop with three constraints: (i) a *small fixed tool set* passed in per call—*evidence_search* over the paper’s text, figures and tables, *figure_parse* for structured figure/table content, and *policy_lookup* for policy metadata such as announcement versus implementation dates; (ii) a *hard step budget* (*max_steps*) so a run cannot wander indefinitely; and (iii) a *logged trace* of every step, tool call, and cited piece of evidence. All other stages are plain orchestration with no agent loop.

Why bound the agency. A fully autonomous agent would make per-run behaviour path-dependent and high-variance, undermining the central evaluation claim of §4: that flaw injection provides *reproducible* local ground truth. Restricting agency to two clearly scoped sub-tasks preserves reproducibility (stable, comparable metrics) and evidence-traceability (every judgement anchored to logged evidence a human can inspect). The expert sits at the end of the pipeline: ARGUS surfaces and localizes identification risk, it does not adjudicate it.

3.3 Output

ARGUS emits (*study*, *identification dimension*, *risk*) triplets, each with an evidence-grounded rationale, alongside the human-readable report. It never produces a verdict on the true causal effect.

This output shape is what makes the system both auditable by a human and measurable by the evaluation that follows.

4 Evaluation by Flaw Injection

4.1 Local ground truth without the counterfactual

The defining difficulty in evaluating a causal-identification auditor is that the quantity of ultimate interest—the true causal effect of a carbon price, an emissions cap, or an emission standard—is never observed. We therefore do not attempt to certify that ARGUS recovers true effects. Instead we construct *local* ground truth by controlled perturbation. Beginning from a credible policy-evaluation paper, we inject exactly one known flaw that targets a single identification dimension, drawn from the flaw taxonomy (*flaw_taxonomy.yaml*). Because we control which dimension we damaged, we know the correct audit outcome for that dimension—without ever needing the true effect. The bottom band of Figure 2 shows the loop: the clean and injected versions are each audited, and three metrics are computed.

- **Detection:** does the injected version raise risk on the targeted dimension above a threshold?
- **False alarm:** does the clean version stay quiet on that dimension?
- **Localization:** is the targeted dimension the one the auditor flags most strongly?

4.2 Realistic perturbation, not sentinels

How a flaw is injected is a methodological choice that matters. A naïve injector appends a sentence that *names* the flaw—e.g. “treated and control have non-zero pre-period coefficients with significant leads.” If the detector then searches for exactly those phrases, detection becomes a tautology: the injector plants the very string the detector greps for. In our initial implementation this produced a perfect-looking but meaningless detection rate of 1.000.

We replace this with *realistic structural perturbation* of two kinds. *Omission* flaws remove the section, figure, or table that supplies the support for a dimension (e.g. deleting the placebo-robustness section or the balance table), modelling

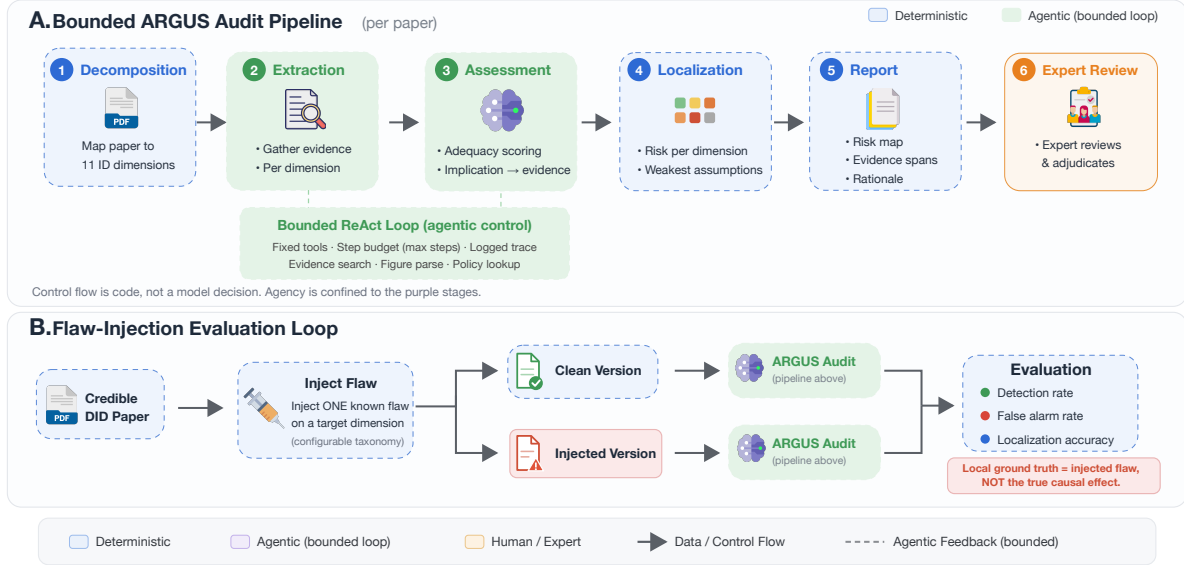


Figure 2: ARGUS architecture. The top band is the audit pipeline, a fixed deterministic sequence; agency is confined to the two agentic stages (extraction and assessment), each a bounded ReAct loop with a fixed tool set (evidence search, figure parse, policy lookup), a hard step budget, and a logged trace. The bottom band shows the flaw-injection evaluation loop, which injects one known flaw, audits the clean and injected versions, and scores detection, false alarm, and localization against the injected flaw as local ground truth—never against the unobserved true effect.

a paper that simply omits the check; an auditor can flag this non-circularly by noticing the *absence* of evidence. *Commission* flaws *rewrite* a section in flawed-but-natural prose, the way a real flawed paper would describe a mistaken design choice (e.g. pooling all province-years in a two-way fixed-effects regression under staggered policy timing), *without* containing any of the phrases the detector looks for. We enforce the latter with a regression test that fails if an injected paper contains the target dimension’s negative-signal phrases, so the circularity cannot silently return. Detection is thereby a genuine task rather than a string match, which is what makes the results in §5 meaningful.

5 Results

We evaluate ARGUS along four axes: a synthetic flaw-injection protocol, an architecture ablation, a run over 27 real difference-in-differences (DID) papers (empirical economics broadly, with environmental policy—carbon pricing, emissions trading, emission standards—as the motivating case), and a calibrated human-gold pilot. Four findings emerge: (i) judging evidence *adequacy*, not keyword presence, is what detects identification flaws; (ii) the bounded architecture—not raw model capability—recovers flaws a single pass misses and turns retrieval failure into an auditable

unknown signal; (iii) on real papers the binding constraint shifts from causal reasoning to evidence grounding; and (iv) against a human gold ARGUS is systematically over-severe, a bias that a lightweight calibration layer reduces while preserving high-risk recall.

5.1 Adequacy beats presence; architecture beats a single pass

We compare three assessors under one flaw-injection protocol on the `clean_supported` fixture (one flaw injected per run, all eleven flaws; `max_steps=1`). A deterministic **keyword baseline** scores each dimension from the presence of supporting and risk-indicating vocabulary. The **two-stage ARGUS assessor** judges evidence *adequacy*—whether the reported evidence actually establishes the assumption—over the *same retrieved evidence* as the keyword baseline, so that pair isolates the judgement policy. A **single-pass** ablation removes the architecture entirely: the full paper and all eleven dimensions in one `gpt-4o` call, with the same model, temperature, risk schema, and scoring harness, but no staged retrieval, relevance gating, bounded tool use, or explicit unknown state. The only difference is architectural. Table 1 reports the headline metrics.

Table 1: Synthetic flaw injection (clean_supported, eleven flaws; all LLM rows gpt-4o): keyword baseline, a same-model *single-pass* ablation, and the *two-stage* ARGUS assessor. Keyword and two-stage share the same retrieved evidence; single-pass shares the model and rubric but removes the architecture. Detection rises from keyword presence, through single-pass adequacy reasoning, to bounded per-dimension auditing. The final row is the leaked, pre-fix circular setup.

Assessor	detection	false alarm	localization
keyword baseline	0.182	0.000	0.182
single-pass (ablation)	0.455	0.000	0.455
two-stage ARGUS	1.000	0.000	0.909
(circular, pre-fix)	1.000	0.000	1.000

The keyword baseline is blind to commission flaws. The baseline never raises a false alarm and correctly flags the two omission flaws (*measurement_break*, *spillover_contamination*) by the *absence* of their vocabulary, but beyond those detection is only 0.182 (Figure 3B). Two failure modes explain it: *cross-section vocabulary bleed*, where deleting the placebo section does not help because another section also mentions “placebo,” and, more fundamentally, blindness to *every* commission flaw, where flawed-but-plausible evidence is present in the expected topic vocabulary. Keyword presence is not evidence adequacy.

Reasoning over adequacy closes the gap. Swapping *only* the assessment policy—extraction unchanged, so both see the same evidence—to an LLM that judges adequacy raises detection from 0.182 to 1.000 with no false alarms and 0.909 localization. The model reads flawed-but-natural prose the keyword scorer cannot: for the injected anticipation flaw it returns *high/flawed*, reasoning that “outcomes began changing between the policy announcement and rollout,” with no sentinel keyword present. This is the empirical core of the design: placing agency where reasoning is required.

The architecture, not the model, does the work. The single-pass ablation isolates the bounded design. On the flaw-injection benchmark it detects only 0.455 of flaws (false alarm 0.000, localization 0.455), against 1.000/0.000/0.909 for the two-stage assessor: judging eleven dimensions from the whole paper at once misses more than half the planted flaws that per-dimension retrieval isolates. On the 25 papers common to both runs (275

cells) the contrast sharpens. Single-pass almost never abstains—its unknown share is 0.4% versus 38.9% for the two-stage pipeline. The cells the bounded pipeline marks unknown become confident judgements under single-pass, mostly *low* (1.8% → 14.9%) or *medium* (14.2% → 43.6%), while the high-risk share is comparable (41.1% vs 45.1%). Without a gate the model commits a substantive risk for nearly every dimension instead of reporting that it could not ground the evidence. The bounded design thus recovers flaws a single pass misses and turns retrieval failure into a visible, auditable signal—the distinction the rest of the section relies on.

Scope. These are synthetic fixtures with few dimensions; 1.000 is a proof-of-concept on clearly injected flaws, not a solved task. The sharper question is whether the behaviour survives on real published papers.

5.2 From synthetic fixtures to real papers: a relocated bottleneck

We next audit a corpus of 27 published DID papers. There is no injected ground truth; we report the raw per-dimension risk distribution of each assessor (Figure 5). The keyword baseline is permissive—91% of its judgements are *low*: it sees topic vocabulary, declares the dimension supported, and flags almost nothing. The two-stage LLM is differentiated, and its most informative output is that ~39% of judgements are *unknown*: the relevance gate could not surface evidence, so the system says so rather than over-flagging. Two regimes appear—dimensions whose discussion is diffuse or lives in figures the text retrieval cannot reach (parallel trends, no anticipation, placebo) are dominated by *unknown*, while dimensions where a section is retrieved (sample period, concurrent policies, SUTVA) are judged inadequate. The finding is methodological: moving to real papers is bottlenecked not by causal reasoning but by *evidence grounding*, which is why the design reports retrieval quality and an explicit *unknown* state distinct from a substantive high-risk judgement.

Judgements are grounded, not pattern-matched. On a trade-and-wages paper ARGUS retrieves the employment-results section and flags *inference* as high because “the paper only mentions heteroskedastic-consistent standard errors and does not provide information on clustering or

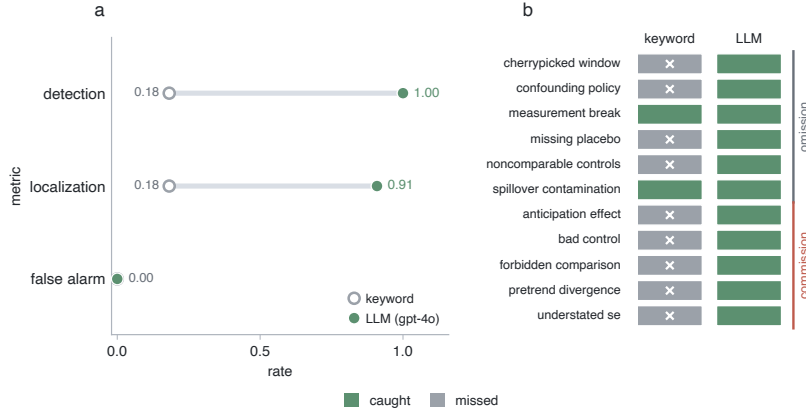


Figure 3: Keyword baseline vs. LLM evidence-adequacy assessor ($n=11$ flaws, realistic injection). **(a)** Swapping *only* the assessment policy—same retrieved evidence—recovers detection $0.182 \rightarrow 1.000$ and localization $0.182 \rightarrow 0.909$; false alarm 0. **(b)** Per-flaw caught/missed: the keyword baseline is *blind to every commission-type flaw*, all of which the LLM catches.

the number of clusters”—a specific econometric observation, not a keyword hit; elsewhere it distinguishes a *related but different* reported check (an alternative-specification section) from the placebo test an assumption actually requires. Roughly 45% of its judgements on these top-journal papers are *high*; without dimension-level expert labels we cannot yet say whether this is warranted or over-strict where retrieval succeeds. We claim the contrast in behaviour and the localization of the grounding bottleneck, not the correctness of any individual flag.

5.3 A calibrated human gold (pilot)

To test whether ARGUS’s flags agree with expert judgement, we built a small expert gold: two annotators independently labelled risk on 5 DID papers $\times$ 11 dimensions (55 cells). Round-1 agreement was low on the raw 3-level scale (Cohen’s $\kappa = 0.17$), but the disagreement is almost entirely ordinal (weighted $\kappa = 0.31$; 96% of disagreements are a single severity step, so annotators almost never reverse direction). A calibration round adjudicated the 27 disagreements: 16 resolved to a single label once the low/medium boundary was anchored, and 11 were flagged intrinsically ambiguous. Those ambiguous cells are not random—they concentrate on *non-canonical DID designs* (continuous-shock, event-study) where a dimension like staggered timing is not applicable as conventionally defined. A real share of “expert disagreement” is thus *design applicability*, not severity, which is itself why even experts abstain. The adjudicated gold (55 cells) is low 29, medium 24,

high 2. This is a pilot ($n=5$, two annotators, one adjudication round), read as preliminary.

5.4 ARGUS versus the human gold

Comparing ARGUS-LLM’s risk to this gold gives the first calibration signal (Table 2, Figure 4). The errors are systematic, not random. ARGUS abstains on 22/55 cells, and when it answers its errors are almost entirely one-directional: 28/33 answered cells are more severe than the adjudicated gold, and none are less severe. Human gold contains only two high-risk cells, whereas ARGUS assigns high risk to 24. Because the pilot gold has so few high labels, high precision and recall are noisy; the robust conclusion is that the current assessor is over-sensitive to missing or partial evidence and is best read as a *high-sensitivity screening tool* for prioritizing human review.

5.5 Calibration: from diagnosis to improvement

The gold comparison also points to a fix. An error analysis of the over-severe cells is sharp: of ARGUS’s over-severe *high* judgements, 23/24 have `evidence_status=missing` and 20/24 have `retrieval_quality=weak`. ARGUS treats “weak retrieval found nothing” as a substantive *high*, conflating a *retrieval failure* with the paper lacking the evidence—which the human, reading the full PDF, resolves to low/medium. This is a fixable calibration target.

A deterministic calibration layer over fields the LLM already produces—the dominant rule demotes a weak-retrieval *high* to *medium* (keep cov-

erage) or unknown (abstain)—cuts over-severity while preserving high-risk recall (0.50): keeping coverage raises exact agreement from 0.12 to 0.45; abstaining reaches weighted $\kappa = 0.25$ but leaves 13 answered cells (Appendix, Table 3, Figure 6). Calibration corrects severity but cannot recover missing evidence—correct *coverage* needs better grounding—so the human gold does more than evaluate ARGUS; it calibrates it.

6 Discussion

What ARGUS measures. A measurement is only as good as the link between the construct and the task that operationalizes it (Jacobs and Wallach, 2021; Raji et al., 2021). ARGUS audits whether a climate-policy study’s identification assumptions are *adequately supported by its reported evidence*: reported-evidence adequacy is an operational proxy for identification credibility, not causal validity itself. The two can diverge—a study may report every diagnostic yet still be invalid, or omit a check yet be sound—so ARGUS surfaces risk for an expert to weigh rather than adjudicating the effect it estimates.

Evidence grounding is the binding constraint. A further gap separates the construct from the task: retrieving the relevant evidence. Our real-paper runs show this, and not the reasoning, is the limiting factor. A reasoning assessor handed off-target evidence over-flags every dimension, so ARGUS reports a *retrieval-quality* status and an explicit *unknown* risk (“the system could not find the evidence”) distinct from a substantive high-risk judgement (“the paper lacks it”). This is why the pipeline separates retrieval, relevance filtering, and adequacy assessment.

Threats to validity. Flaw injection certifies an auditor against the *injected* threat on a single dimension—not against un-injected threats, and it inherits the flaw taxonomy’s blind spots—so we treat detection and localization as necessary but not sufficient, pairing them with the false-alarm rate and the human-gold comparison. Further limitations are stated below.

Future work. Four directions follow from §5: (1) harden the reasoning assessor on subtler flaws, studying calibration, cross-model robustness, and cited-evidence traceability under the same flaw-injection protocol; (2) collect *dimension-level* expert annotation (the real corpus is labelled

per method/claim, not per assumption) to test whether ARGUS’s judgements match expert opinion; (3) run human-expert studies of whether ARGUS reports actually help (trust, time, errors avoided), since the system assists rather than replaces expert judgement; and (4) extend beyond the China emissions-trading case to broader environmental and empirical-economics settings, testing how far the rubric-plus-injection methodology generalizes.

7 Conclusion

The right target for AI assistance in empirical climate-policy research is not the unobservable causal effect but the *identification credibility* of the study that estimates it. ARGUS operationalizes this stance: a bounded-agentic pipeline decomposes identification into auditable dimensions, gathers and assesses in-paper evidence, and localizes risk in a transparent, human-reviewable report, while confining model agency to two scoped, traceable sub-tasks. We validate such auditors with synthetic flaw injection, which supplies reproducible local ground truth without access to the true effect. Once the leakage that made this evaluation circular is removed, a keyword baseline is blind to flawed-but-plausible evidence (0.182 detection), while a model that judges evidence adequacy recovers it (1.000, no false alarms). A pilot human-gold study further shows that expert labels can do more than evaluate the auditor: they expose a systematic over-severity bias and calibrate the mapping from evidence gaps to audit risk. Scaling this from synthetic fixtures to a real, expert-annotated corpus with humans in the loop remains the central open challenge. Auditing identification credibility is, we argue, both a tractable and a responsible role for AI in empirical environmental-policy research.

Limitations

Our human gold is pilot-scale: 5 papers and 55 cells with two annotators and a single adjudication round, so the expert comparison is a preliminary signal rather than a calibrated benchmark. The current scope is confined to difference-in-differences identification; whether the rubric-plus-injection methodology transfers to other quasi-experimental designs is untested. Evidence grounding, not reasoning, is the binding constraint: when retrieval returns off-target pas-

sages the assessor cannot judge adequacy, which is why ARGUS emits an explicit `unknown` state rather than a substantive verdict. Finally, ARGUS is an auditor, not an oracle—it screens whether reported evidence supports a study’s identification assumptions and does not adjudicate the true causal effect, which is never observed. Its outputs are meant to surface risk for expert review, not to replace it.

References

Marianne Bertrand, Esther Duflo, and Sendhil Mullainathan. 2004. How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119(1):249–275.

Rotem Botvinik-Nezer, Felix Holzmeister, Colin F. Camerer, et al. 2020. Variability in the analysis of a single neuroimaging dataset by many teams. *Nature*, 582(7810):84–88.

Abel Brodeur, Nikolai Cook, and Anthony Heyes. 2020. Methods matter: P-hacking and publication bias in causal analysis in economics. *American Economic Review*, 110(11):3634–3660.

Brantly Callaway and Pedro H. C. Sant’Anna. 2021. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

Clément de Chaisemartin and Xavier D’Haultfœuille. 2020. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.

Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92.

Andrew Goodman-Bacon. 2021. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.

Abigail Z. Jacobs and Hanna Wallach. 2021. Measurement and fairness. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*.

Zhijing Jin, Jiarui Liu, Zhiheng Lyu, Spencer Poff, Mrinmaya Sachan, Rada Mihalcea, Mona Diab, and Bernhard Schölkopf. 2024. Can large language models infer causation from correlation? In *International Conference on Learning Representations (ICLR)*.

Zhijing Jin et al. 2023. CLadder: Assessing causal reasoning in language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Emre Kiciman, Robert Ness, Amit Sharma, and Chenhao Tan. 2024. Causal reasoning and large language models: Opening a new frontier for causality. *Transactions on Machine Learning Research (TMLR)*.

Douwe Kiela, Max Bartolo, Yixin Nie, et al. 2021. Dynabench: Rethinking benchmarking in NLP. In *NAACL-HLT*.

Donggyu Lee, Hyeok Yun, Meeyoung Cha, Sungwon Park, Sangyoon Park, and Jihee Kim. 2025. EconCausal: A context-aware causal reasoning benchmark for large language models in social science. Preprint, arXiv:2510.07231. Preprint, arXiv:2510.07231.

Percy Liang, Rishi Bommasani, Tony Lee, et al. 2023. Holistic evaluation of language models. *Transactions on Machine Learning Research (TMLR)*.

Inioluwa Deborah Raji, Emily M. Bender, Amanda-lynn Paullada, Emily Denton, and Alex Hanna. 2021. AI and the everything in the whole wide world benchmark. In *NeurIPS Datasets and Benchmarks Track*.

Jonathan Roth, Pedro H. C. Sant’Anna, Alyssa Bilinski, and John Poe. 2023. What’s trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218–2244.

Ayush Sawarni, Jiayuan Tan, and Vasilis Syrgkanis. 2026. CausalReasoningBenchmark: A real-world benchmark for disentangled evaluation of causal identification and estimation. Preprint, arXiv:2602.20571. Preprint, arXiv:2602.20571.

Raphael Silberzahn, Eric L. Uhlmann, Daniel P. Martin, et al. 2018. Many analysts, one data set: Making transparent how variations in analytic choices affect results. *Advances in Methods and Practices in Psychological Science*, 1(3):337–356.

Liyang Sun and Sarah Abraham. 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.

Linying Yang, Vik Shirvaikar, Oscar Clivio, and Fabian Falck. 2024. A critical review of causal reasoning benchmarks for large language models. Preprint, arXiv:2407.08029. Preprint, arXiv:2407.08029.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Matej Zečević, Moritz Willig, Devendra Singh Dhami, and Kristian Kersting. 2023. Causal parrots: Large language models may talk causality but are not causal. *Transactions on Machine Learning Research (TMLR)*.

Table 2: ARGUS-LLM vs adjudicated human gold (55 cells, 5 papers) under two unknown policies: coverage (drop abstentions) and strict (unknown counts as mismatch). Pilot-scale—only 2 gold cells are *high*; the robust signal is the one-directional over-severity and the abstention rate.

unknown policy	n	exact	κ	wt. κ
exclude (coverage)	33	0.152	0.019	0.044
mismatch (strict)	55	0.091	0.002	—

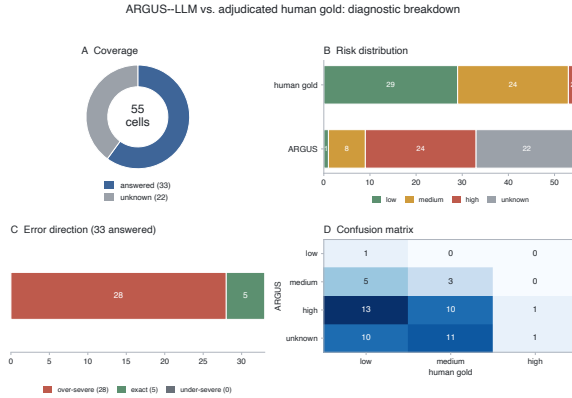


Figure 4: ARGUS-LLM versus adjudicated human gold on 55 dimension-level cells (5 papers). (a) coverage, separating answered judgments from abstentions; (b) severity distributions of human gold and ARGUS; (c) error direction among the 33 answered cells; (d) the ARGUS-by-gold confusion matrix. The dominant signal is one-directional over-severity: ARGUS assigns higher risk than gold in 28/33 answered cases and lower risk in none.

Yu Zhou, Xingyu Wu, Beicheng Huang, Jibin Wu, Liang Feng, and Kay Chen Tan. 2024. [CausalBench: A comprehensive benchmark for causal learning capability of large language models](#). Preprint, arXiv:2404.06349. Preprint, arXiv:2404.06349.

A Supplementary Figures and Tables

This appendix collects supporting material referenced from the main text: the per-dimension risk distribution on the 27-paper real corpus (Figure 5), and the deterministic calibration results against the adjudicated human gold (Table 3, Figure 6), and the agreement metrics for ARGUS versus the human gold (Table 2).

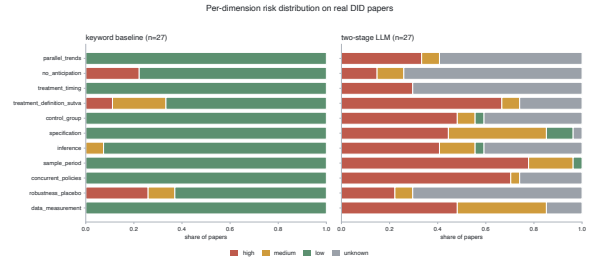


Figure 5: Per-dimension risk distribution on 27 real DID papers. The keyword baseline (left) is almost entirely *low*; the two-stage LLM (right) is differentiated, with a large *unknown* share that marks retrieval failure rather than over-flagging.

Table 3: Deterministic calibration vs the human gold (55 cells), two policies for weak-retrieval *high*: demote to *medium* (keep coverage) or abstain to *unknown*. Both cut over-severity and preserve high recall. Pilot scale—only 2 gold cells are *high*.

metric	before	demote→med	abstain→unk
over-severe (answ.)	29	18	6
high prec. / rec.	0.04 / 0.50	0.50 / 0.50	0.50 / 0.50
exact agr. (answ.)	0.12	0.45	0.54
wt. κ (answ.)	0.06	0.13	0.25
answered / unk	33 / 22	33 / 22	13 / 42

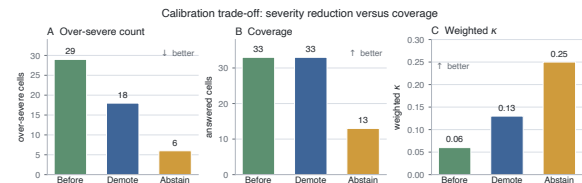


Figure 6: **Calibration trade-off: severity reduction versus coverage** against the adjudicated human gold. Demoting weak-retrieval *high*s to *medium* preserves coverage (33 answered) while cutting over-severe judgments from 29 to 18; abstaining gives the strongest correction (over-severe 6, weighted κ 0.25) but leaves only 13 answered cells. Calibration corrects severity bias; the remaining bottleneck is evidence grounding, not severity thresholding alone.